Update Conference
PRAGUE 2026

Code that survives the crash

Durable Task Scheduler for workflows and AI agents

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Hi, I'm Tamir Dresher

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Flight booked. Hotel booked. No car.

1

Flight
booked

1

Hotel
booked

0

Car
not booked



The process died before the car booking.

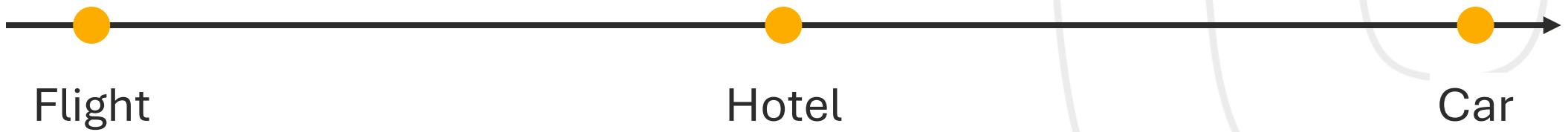
What got booked? Will a rerun book twice?

Three awaits. Who remembers the progress?

NaiveTrip.BookAsync(booking, trip, attemptId) · new attemptId per run

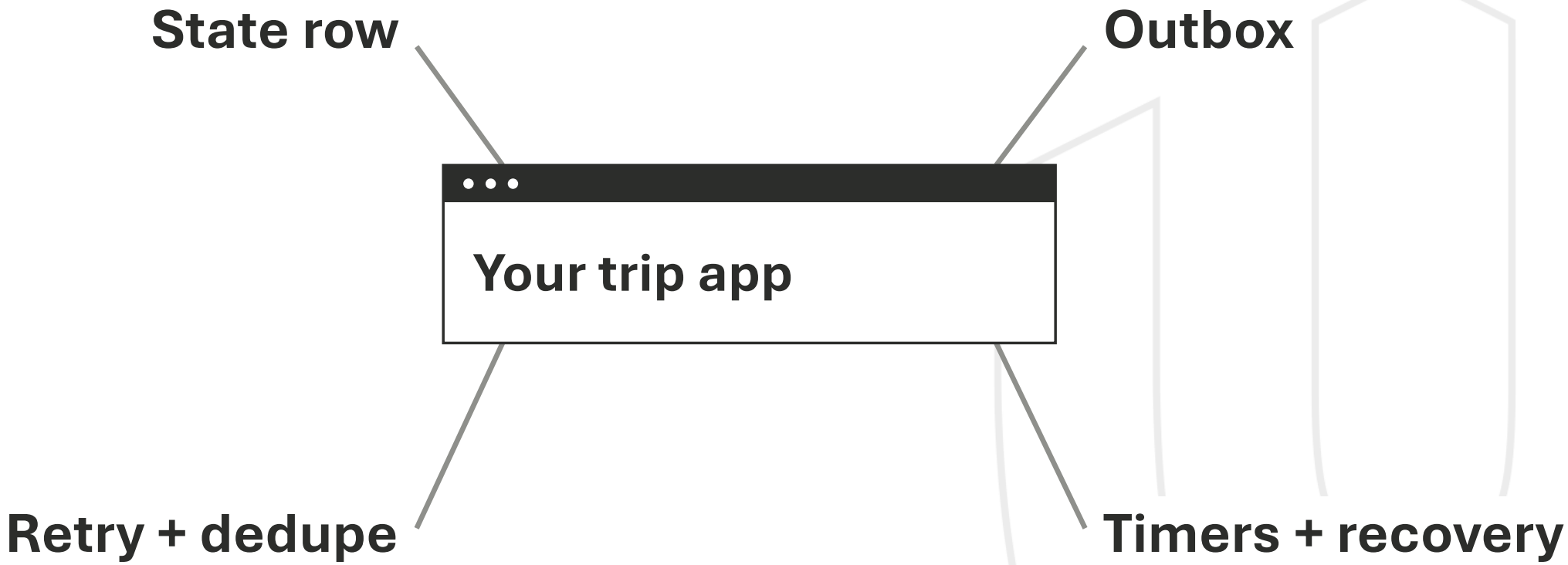
LIVE DEMO

```
BookingReceipt flight = await booking.BookFlightAsync(
    new($"{attemptId}/flight", "flight", trip));
BookingReceipt hotel = await booking.BookHotelAsync(
    new($"{attemptId}/hotel", "hotel", trip));
BookingReceipt car = await booking.BookCarAsync(
    new($"{attemptId}/car", "car", trip));
```



Rerun: 2 flights · 2 hotels · 1 car

Your trip app just hired a workflow team.



Congratulations. You also own a workflow engine.

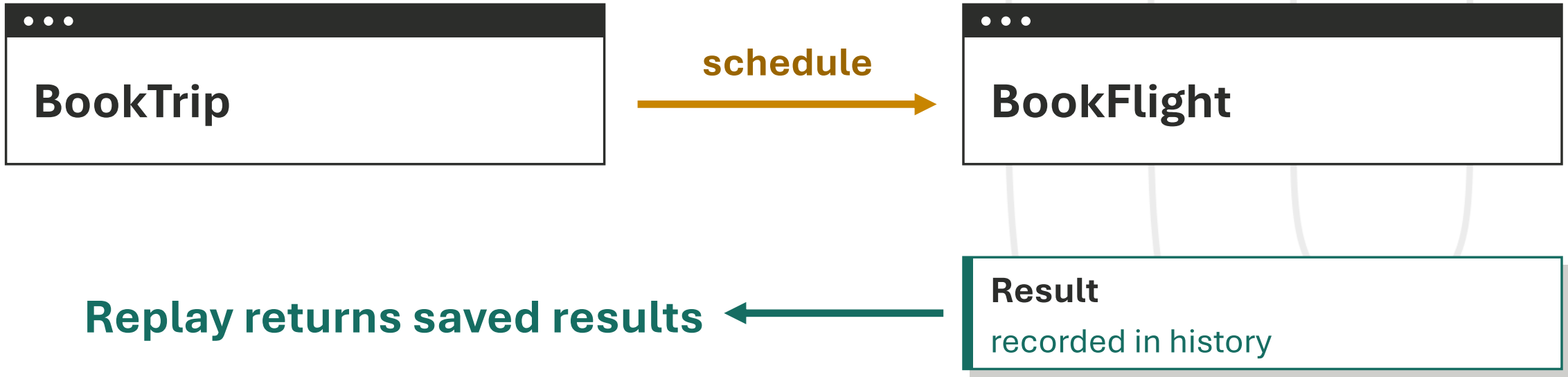
Keep the code. Give the trip a durable memory.

TaskOrchestrator

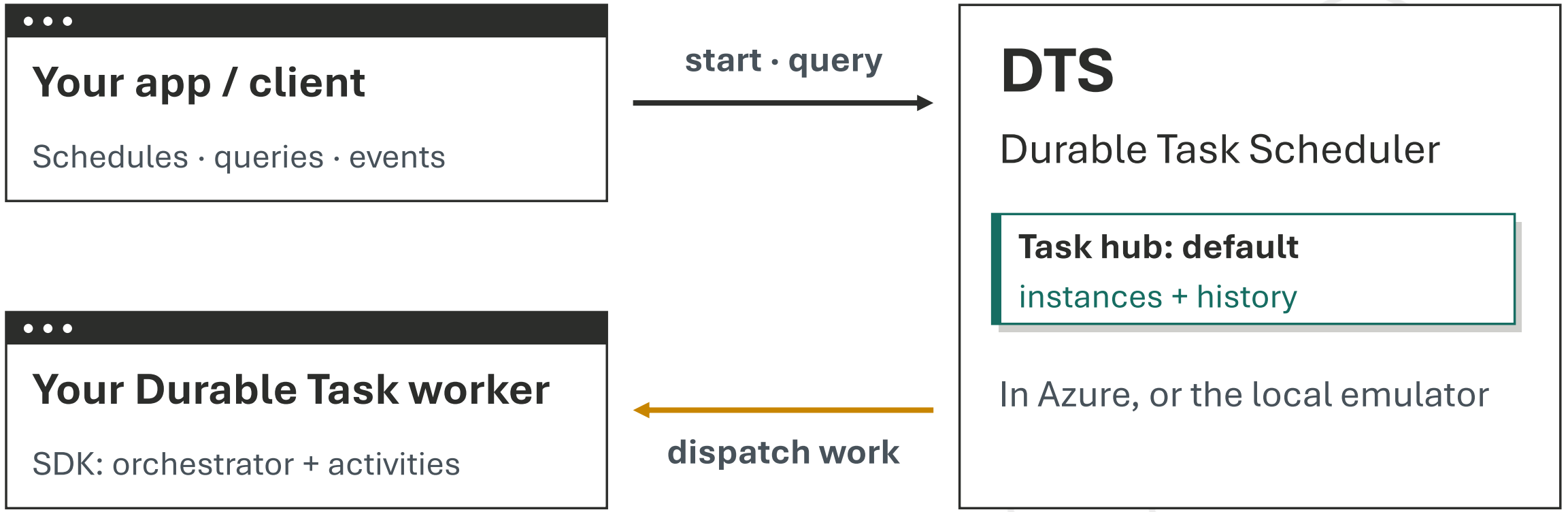
Decide the next step

TaskActivity

Execute I/O



Your worker runs C#. DTS keeps the history.



DTS does not host your C# methods.

The flight await creates a recorded boundary.

BookTrip.RunAsync(context, trip) WORKER RUNNING

```
BookingReceipt flight = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookFlight), trip);
BookingReceipt hotel = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookHotel), trip);
BookingReceipt car = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookCar), trip);
```

DURABLE HISTORY

BookFlight
scheduled

New command → dispatch the activity.

The flight receipt lands in history.

`BookTrip.RunAsync(context, trip)` **WORKER RUNNING**

```
BookingReceipt flight = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookFlight), trip);
BookingReceipt hotel = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookHotel), trip);
BookingReceipt car = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookCar), trip);
```

DURABLE HISTORY

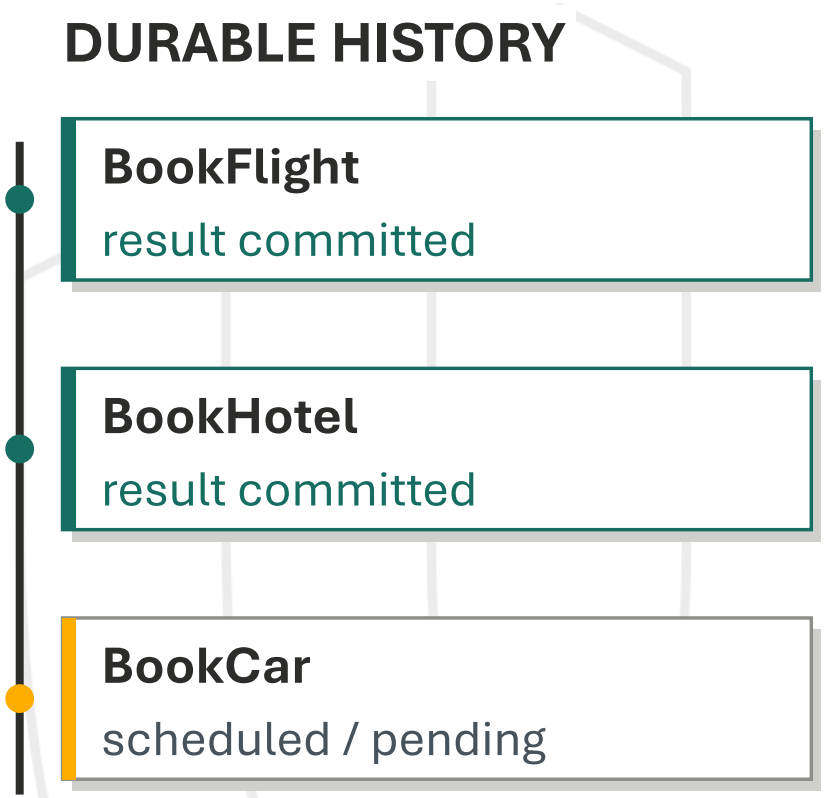
BookFlight
result committed

Completion + result are committed.

Hotel recorded. Car still unbooked.

BookTrip.RunAsync(context, trip) WORKER RUNNING

```
BookingReceipt flight = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookFlight), trip);
BookingReceipt hotel = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookHotel), trip);
BookingReceipt car = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookCar), trip);
```



Flight + hotel recorded. Car still 0.

Kill the worker. Keep the two receipts.

`BookTrip.RunAsync(context, trip)` **WORKER DOWN**

```
BookingReceipt flight = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookFlight), trip);
BookingReceipt hotel = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookHotel), trip);
BookingReceipt car = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookCar), trip);
```

DURABLE HISTORY

- **BookFlight** — result committed
- **BookHotel** — result committed
- **BookCar** — scheduled / pending

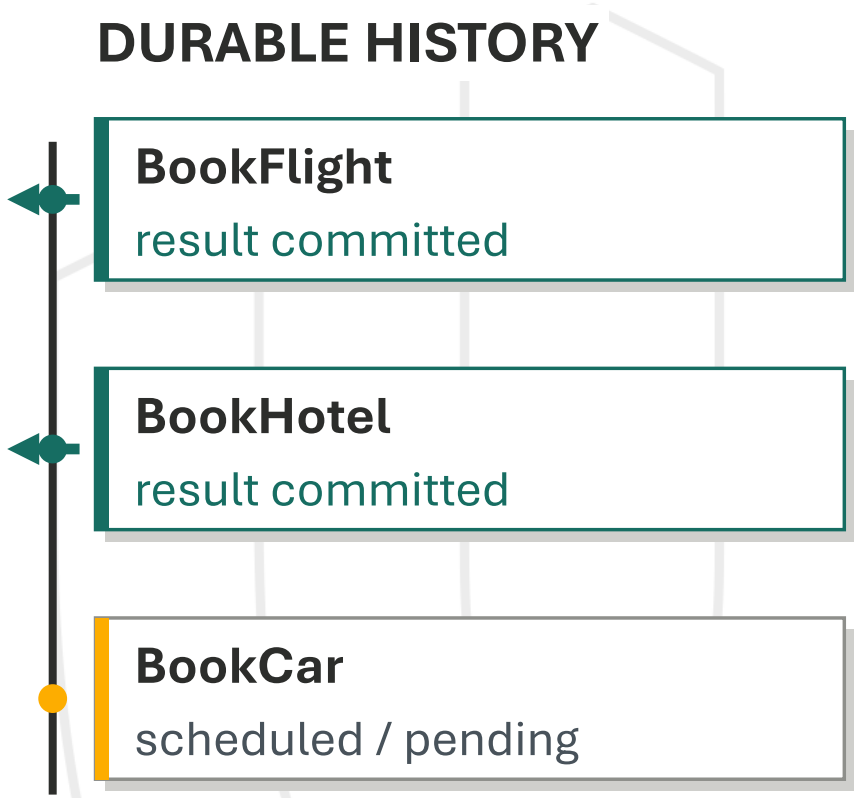
Stack lost. Two committed results retained.

Restart at line one. Return the saved results.

`BookTrip.RunAsync(context, trip)` **NEW WORKER**

```
BookingReceipt flight = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookFlight), trip);
BookingReceipt hotel = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookHotel), trip);
BookingReceipt car = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookCar), trip);
```

Flight + hotel came back from history



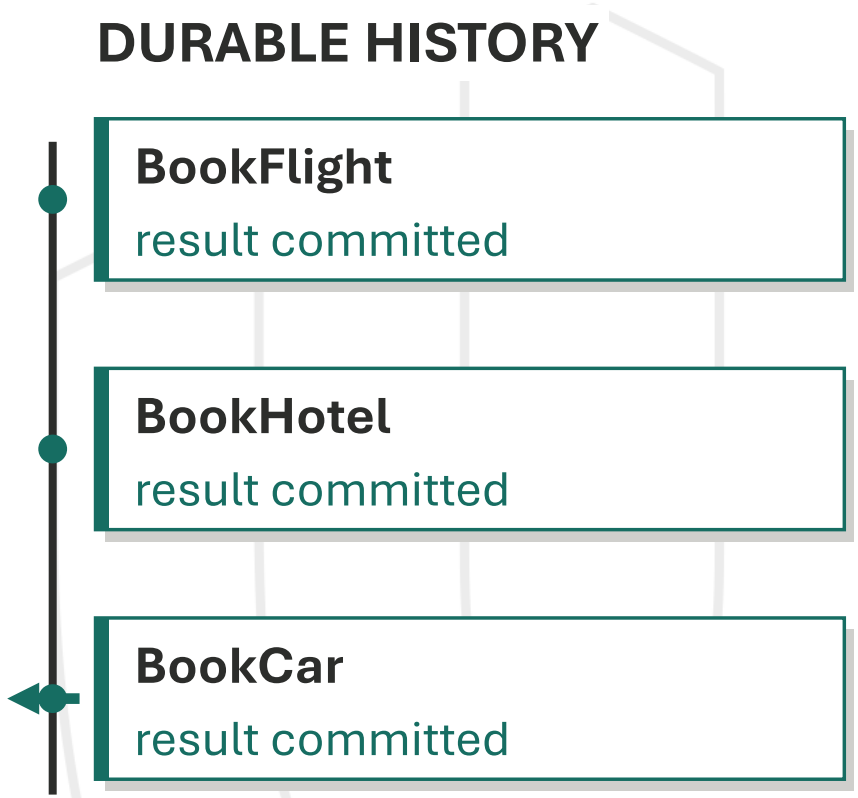
Code runs again. Completed bookings aren't reissued.

Continue the car. Finish 1 / 1 / 1.

`BookTrip.RunAsync(context, trip)` **NEW WORKER**

```
BookingReceipt flight = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookFlight), trip);
BookingReceipt hotel = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookHotel), trip);
BookingReceipt car = await
  context.CallActivityAsync<BookingReceipt>(
    nameof(BookCar), trip);
```

The car runs for real: 1 / 1 / 1



Same instance. Three bookings. One of each.

Pull the plug between hotel and car.

LIVE DEMO

Resume the trip. Not a new trip.

1

Same instance ID

2

Two saved results

3

1/1/0 → 1/1/1

Watch those three things, then the dashboard timeline.

Will the same trip choose the same next call?

Inside BookTrip.RunAsync(context, trip):

```
if (DateTime.Now.Hour < 17)
    await context.CallActivityAsync<BookingReceipt>(
        nameof(BookFlight), trip);
else
    await context.CallActivityAsync<BookingReceipt>(
        nameof(BookHotel), trip);
```

FIRST RUN · 16:59

History: BookFlight

REPLAY · 17:01

Code: BookHotel

Same code. Different branch. History and command no longer match.

Replay needs the same decisions.

Inside the orchestrator:

Replay-safe: use the context

```
DateTime created = context.CurrentUtcDateTime;
Guid bookingId = context.NewGuid();
```

Not replay-safe: move it into an activity

DateTime.Now · Guid.NewGuid() · Random · HTTP · database / file I/O

Replay reconstructs decisions—not arbitrary side effects.

Hotel: two attempts. One reservation.

Hotel commits

provider owns receipt

ACK is lost

worker crashes

Hotel repeats

same operation key

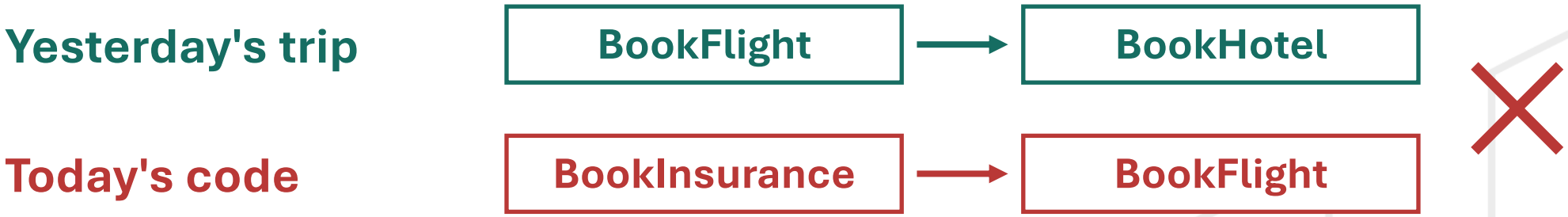


Inside BookTrip.RunAsync(context, trip), the hotel call passes:

```
new BookingCommand(
    $"{context.InstanceId}/hotel", "hotel", trip)
```

“Exactly once” = saved result + stable key + provider dedupe.

Add insurance today. Don't break yesterday's trips.



Client registration: new trips start as version 2.0

```
client.UseDefaultVersion("2.0");
```

Inside BookTrip.RunAsync(context, trip): pre-upgrade trips skip the new step

```
if (context.CompareVersionTo("2.0") >= 0)
    await context.CallActivityAsync<BookingReceipt>(
        nameof(BookInsurance), trip);
```

Old trips replay the old path. New trips get insurance.

Retry the flight call, not the entire trip.

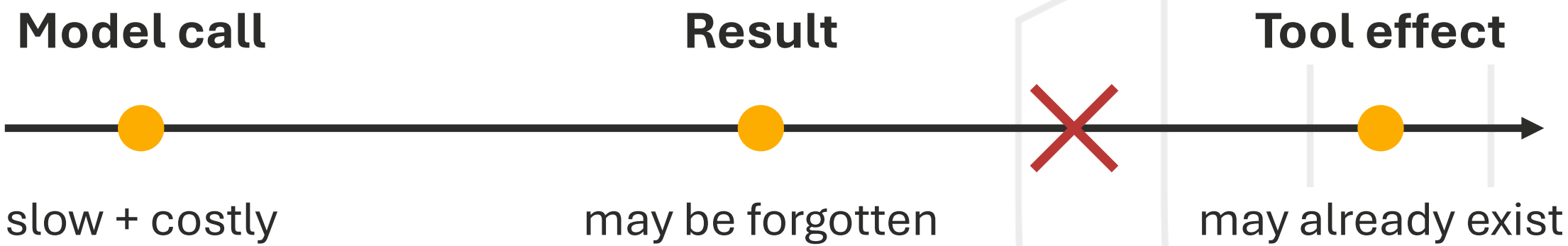
Inside BookTrip.RunAsync(context, trip):

```
BookingReceipt flight = await
    context.CallActivityAsync<BookingReceipt>(
        nameof(BookFlight),
        new BookingCommand(
            $"{context.InstanceId}/flight", "flight", trip),
        TaskOptions.FromRetryPolicy(
            new RetryPolicy(3, TimeSpan.FromSeconds(1))));
```



Retry the work. Don't multiply the business effect.

Agents make forgotten work more expensive.



Conversation memory is not workflow recovery.

Swap the provider, not the workflow.

IChatClient → AsAIAgent → AddAIAgent

Inside ConfigureDurableAgents(options => ...); chatClient is any IChatClient:

```
var planner = chatClient.AsAIAgent(
    instructions: "Plan a trip within the budget.",
    name: "TripPlanner");
options.AddAIAgent(planner);
```

Demo model is scripted. Swap the IChatClient; the workflow stays.

Planner output becomes the reviewer's input.

Inside PlanAndBookTrip.RunAsync(context, input):

```
DurableAIAgent planner = context.GetAgent("TripPlanner");
AgentSession plannerSession =
    await planner.CreateSessionAsync();
AgentResponse itinerary = await planner.RunAsync(
    JsonSerializer.Serialize(input.Trip), plannerSession);
DurableAIAgent reviewer = context.GetAgent("BudgetReviewer");
AgentSession reviewerSession =
    await reviewer.CreateSessionAsync();
AgentResponse review = await reviewer.RunAsync(
    itinerary.Text, reviewerSession);
```

TripPlanner → BudgetReviewer → immutable itinerary

Hold a deadline, not a worker thread.

Inside PlanAndBookTrip.RunAsync(context, input):

```
DateTime deadline = context.CurrentUtcDateTime.AddMinutes(15);
using CancellationTokenSource timerCancellation = new();
Task timeout = context.CreateTimer(deadline, timerCancellation.Token);
Task<ApprovalSignal> approval =
    context.WaitForExternalEvent<ApprovalSignal>("TripApproval");
if (await Task.WhenAny(approval, timeout) == timeout)
    return "TimedOut";
timerCancellation.Cancel();
```



hold / deadline



approval event

Wake-up ≠ authorization. Cancel the losing wait.

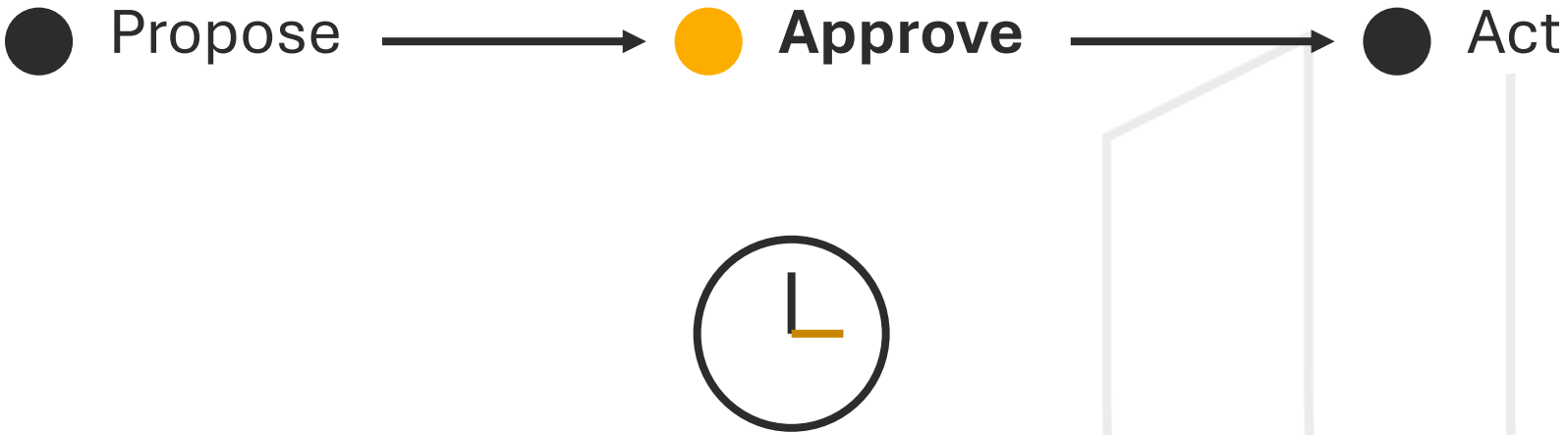
Crash the agents. Don't pay for the planner twice.

LIVE DEMO · Kill the worker while it waits for the traveler



Saved results come back from history. A call that was mid-flight runs again.

A model can propose. A traveler must authorize.



Frozen plan + deadline + trusted validation

A second model's agreement is not authorization.

Choose the boundary you want to own.

	BEST WHEN	YOU STILL OWN
Hangfire	Background jobs and continuations	Job code, storage, workers and retry-safe effects
MassTransit saga	Correlated messages and explicit process state	State machine, persistence and message contracts
Temporal	Code-first workflows on Temporal's runtime	Workflow/activity code, workers, replay compatibility
Durable Functions	Workflows hosted in Azure Functions	Function code, hosting plan and backend configuration
Durable Task + DTS	Code-first workflows on your worker compute	Worker host, workflow code, hub access (RBAC), safe effects

Sometimes a queue is exactly enough.

One short idempotent job	✓	**Queue / Hangfire**
One atomic data change	✓	**Database transaction**
An explicit message-driven process	✓	**Keep the saga**

No durable coordination requirement? Don't add one.

Same trip. Same crash. In Python.

From python/book_trip.py (durabletask-python, same emulator):

```python
def book_trip(ctx: task.OrchestrationContext, trip: dict):
    flight = yield ctx.call_activity(
        book_flight, input=f"{ctx.instance_id}/flight")
    hotel = yield ctx.call_activity(
        book_hotel, input=f"{ctx.instance_id}/hotel")
    car = yield ctx.call_activity(
        book_car, input=f"{ctx.instance_id}/car")
    return {"trip": trip, "bookings": [flight, hotel, car]}
```

Killed before the car, new worker: flight 1 · hotel 1 · car 1

Same crash, same resume. Python yields where C# awaits.

Test the failure boundaries before the happy path.

1 **Crash at the boundary**

Can you explain every receipt?

2 **Repeat one effect**

Can the owner deduplicate it?

3 **Upgrade a waiting instance**

Does yesterday's run still finish?

Start with one workflow you cannot afford to forget.

- What must survive?
- What may repeat?
- Who owns the wait?



Slides + code

github.com/tamirdresher/DurableTaskUpdatePrague2026

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Thank you!



@tamir_dresher in/tamirdresher tamirdresher

#UCP26

Session: 26

GIVE US FEEDBACK FOR THIS SESSION



Code That Survives the Crash: Durable Task Scheduler for Workflows and AI Agents